

Haorui (Rae) Wang

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Education

University of Michigan – Ann Arbor

Aug. 2023 – May 2027

B.S. in Data Science & B.S. in Statistics

GPA: 3.7/4.0, University Honors

Relevant Coursework: Data Mining, Statistical Computing, Applied Regression, Database Management, Data Structures & Algorithms

Professional Experience

Research Assistant – LLM-Based Clinical Extraction - University of Michigan

Jan 2026 – Present

- Built a production-style **LLM generator-evaluator pipeline** transforming free-text radiology reports into structured patient timelines for downstream survival modeling.
- Designed **self-healing validation loops** (generate → validate → repair) to automatically detect and correct LLM hallucinations and formatting failures at scale.
- Trained transformer-based survival models with Cox proportional hazards loss; developed evaluation pipelines using concordance index (C-index) and ranking-based metrics.

Student Researcher – Healthcare & Clinical Analytics - UofM & Trinity Health

Jan 2025 – Dec 2025

- Processed **200K+ medical billing records and clinical notes**; engineered NLP features (TF-IDF, Word2Vec) from unstructured text to support model training.
- Built end-to-end ML classification pipelines (XGBoost, RF, SVM, LR) achieving **82% F1-score**; integrated results into production analytics workflows reducing manual review time by **30%**.
- Developed LLM-assisted anomaly detection workflows using Azure OpenAI + Microsoft Fabric, deploying data-driven insights to business stakeholders.

Research Assistant – ML/DL for Climate Forecasting - University of Michigan

Aug 2024 – May 2025

- Trained climate forecasting models (ConvGNP) on GPU-based HPC clusters using **10+ years** of archived data; implemented time-varying and spatially-varying bias-correction techniques.
- Visualized loss curves and error distributions; confirmed smoother convergence and improved forecast reliability across regional climate metrics.

Data Analysis Intern – Haier Global Business Co., Ltd.

Jun 2024 – Aug 2024

- Analyzed **500K+ Amazon sales records** to uncover high-growth segments and price inefficiencies; automated weekly reporting pipelines cutting manual analysis time by **40%**.
- Delivered price elasticity forecasts adopted by product managers, driving **58% YoY sales growth**.

Projects

FitForecast – Personalized Behavioral Analytics Platform

Jan 2026 – Present

- Built a full-stack product analytics platform: users log workouts and meals in natural language; system extracts behavioral patterns via rule-based NLP parsing and rolling 14/30-day baseline models.
- Designed a **deterministic insight engine** surfacing statistically-grounded, explainable behavioral trends relative to each user's personal baseline – analogous to product metric alerting systems.
- Developed end-to-end pipelines from data ingestion and feature extraction through to trend visualization.

Alfred – AI Workflow Agent System

Feb 2026 – Present

- Architected a **human-in-the-loop multi-agent platform** converting natural language into executable workflows with permission control and action previews for safe real-world execution.
- Integrated **16 tool APIs** (email, calendar, search, browser automation, messaging) into a modular planner + execution engine; implemented XP leveling and engagement mechanics for long-term retention.

Technical Skills

Programming: Python, SQL, R, C++, TypeScript

ML / Data Science: PyTorch, TensorFlow, scikit-learn, NLP, LLMs, Survival Analysis, Statistical Modeling

AI Systems: Multi-Agent Systems, LLM Pipelines, Prompt Engineering, Transformers, Deep Learning

Data & Analytics: pandas, NumPy, PySpark, Matplotlib, Data Visualization, A/B Testing Concepts

Cloud & Infra: AWS, Azure, Microsoft Fabric, High Performance Computing

Certifications: IBM Machine Learning Professional Certificate; Google Data Analytics Professional Certificate